

Event-Time MCDM Allocation across DeFi Lending Protocols: An HFT-Inspired Methodology with Walk-Forward Validation

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Abstract

We design an event-time, gas-aware multi-protocol allocator for USDC supply markets across the six largest Ethereum-L1 lending protocols (Aave V3, Compound V3, Spark, Morpho Blue, Euler V2, Fluid; $\sim 67\%$ of $\sim \$54\text{B}$ TVL). A three-tier policy ladder is evaluated on every block: T1 gas-aware threshold, T2 OU optimal stopping with closed-form Bellman boundary, T3 Cox proportional-hazards on MacKenzie (2021) Table 3.2 F1/F3/F4 signal-class features trained with López de Prado AFML methodology (triple-barrier labels, purged k -fold CV with embargo, sample-uniqueness weighting). Over six non-overlapping three-month walk-forward windows (Nov 2024 – Apr 2026), T3 beats passive holds of all six protocols on the 6-way active panel: +2.88 pp vs Aave, +2.70 pp vs Compound, +1.85 pp vs Spark, +2.65 pp vs Morpho, +2.70 pp vs Fluid, +1.53 pp vs Euler (paired-bootstrap $p < 0.05$ on all six; 5–6 of 6 windows positive). The Cox model adds +7.03 bp over T1 ($p = 0.015$), closing pre-registered H1c. The Maker DSR one-hour delta and USDC peg deviation are the strongest non-fragmentation predictors. A production-grade Python agent (Flashbots private mempool, Prometheus observability) shares decision modules bit-identically with the backtest.

Keywords: ICICPE, DeFi, MCDM, BiGRU-CNN, cross-domain transfer

1. Introduction

Multi-protocol DeFi lending allocation has been studied either as a rate-forecasting problem on hourly-aggregated rate series [1] or as a control-theoretic problem with reinforcement learning on the protocol-design side [2, 3, 4]. Both formulations share an implicit assumption: that hourly resolution is sufficient for an allocator to extract economically meaningful edge. We argue that this assumption is empirically wrong. Lending-rate crossovers between Ethereum L1 protocols happen at the speed of pool deposits and withdrawals — per-block ($\sim 12\text{ s}$ post-PoS), not per-hour — because the rate is a deterministic function of realised utilization, and utilization updates exactly once per on-chain interaction [5]. Aggregating to hourly bars destroys the inter-block microstructure where the allocator’s competitive edge actually lives.

Thesis. We argue that DeFi lending allocation is structurally analogous to high-frequency trading microstructure as described by [7], and that the four-class HFT signal taxonomy of that work transposes onto the multi-protocol lending problem with a clean mapping. Concretely: the cross-

protocol rate spread is itself the decision variable (the dominant signal); mempool-visible transactions are the analog of “order-book dynamics”; lead rates from Maker DSR and Curve 3pool stand in for the “futures lead” class; and the on-chain gas regime plus stablecoin peg deviations play the role of “related instruments”. Under this lens the natural decision frame is event-time, gas-aware, with three nested policy tiers — gas-aware threshold (T1), Ornstein–Uhlenbeck optimal stopping with switching cost (T2), and Cox proportional-hazards regression on the signal vector (T3) — each grounded in a separate microstructure or quantitative-finance source book [8, 9, 10, 11, 7]. This formulation extends the published 4-factor MCDM baseline of [12] from hourly cliff-edge switching to per-block gas-aware switching, and from a 2-way (Aave + Compound) panel to the 6-protocol matrix that covers $\sim 67\%$ of total Ethereum-L1 USDC-lending TVL.

Empirical finding. On a 3,931,200 per-block panel covering January 2026–April 2026 across 3 Ethereum USDC lending venues (Aave V3, Morpho Blue, Euler V2) plus three passive-hold benchmarks (Compound V3, Spark, Fluid), T1

beats passive Aave V3 hold under paired monthly Sharpe bootstrap ($\Delta\text{SHARPE} = +5.048$, 95 % CI $[+2.691, +17.899]$, $p = 0.011$); the sophisticated-retrain T3 (Cox PH on the F1+F3 design matrix) closes pre-registered H1c with $\Delta\text{APY} = +7.03$ bp over T1 ($p = 0.015$) on the $N=6$ walk-forward paired bootstrap. Two orders of magnitude more rebalance opportunities than the hourly baseline of [12] expresses.

Contributions.

- **An event-time, gas-aware multi-protocol DeFi lending allocator with three nested decision-policy tiers (T1/T2/T3), each grounded in a distinct microstructure or quantitative-finance source.** We give the closed-form gas-cost crossover inequality that locks T1’s threshold, the OU–Bellman analytical boundary that closes T2, and the Cox proportional-hazards specification with the triple-barrier label of [11] for T3.
- **The four-class HFT signal taxonomy of [7] (Table 3.2) operationalised on DeFi lending.** F1 (lead) $\leftarrow$ Maker DSR rate plus lagged spreads to the top-protocol APR; F3 (fragmentation) $\leftarrow$ the cross-protocol rate spread itself, the dominant signal and the F3-only T3 fit’s sole driver; F4 (related) $\leftarrow$ ETH/USD, gas regime, USDC peg deviations; F2 (order-book dynamics) deferred per the explicit risk register pending paid mempool coverage.
- **López de Prado AFML methodology applied to DeFi lending:** triple-barrier labels (24-h horizon), purged k -fold CV with embargo ≈ 5.4 days per [11, Ch. 7], sample-uniqueness weighting, Deflated Sharpe Ratio gate [11, Ch. 14.7.3].
- **The first per-block-resolution head-to-head matrix of seven allocation policies on a multi-month Ethereum mainnet panel** (four baselines plus T1/T2/T3), with walk-forward $N \times M$ paired bootstrap as the binding inference and a regime-conditional Q1/Q2 breakdown that exposes which tier dominates which volatility regime.
- **A production-grade reference agent + full reproducibility envelope.** The same decision/ Python package is consumed *bit-identically* by an off-chain agent (per-block newHeads subscription, Flashbots pri-

vate mempool — the asymmetric speed-bump of [7, pp.200–203]; [13] survey the defence space). The 6-way panel ($\sim 3.9\text{M}$ rows), Cox sidecars, walk-forward equity parquets, and Institutional Dossier are public on HuggingFace + GitHub.

Outline. Section 2. reviews DeFi lending mechanics, the HFT canon we borrow from, and recent DeFi-microstructure / MEV work [14, 15]. Section 3. specifies the three-tier ladder. Section 4. reports the seven-policy matrix and the binding walk-forward bootstrap. Section 5. positions the result against the HFT canon and the cross-domain hinge; Sections 6. and 7. state scope and next steps.

2. Background and Related Work

We synthesize four strands of literature that bracket the contribution: DeFi lending mechanics, the HFT microstructure canon we borrow from, MCDM in algorithmic allocation, and the equilibrium impossibility result of [16] as the most relevant theoretical counter-evidence.

2.1 DeFi lending mechanics

Aave V3, Compound V3, Morpho Blue, Euler V2, Spark, and Fluid all implement deterministic interest-rate models in which the supply APY is a piecewise-linear (or in Morpho Blue’s case, adaptive) function $f_{\text{kink}}(u)$ of pool utilization $u = \text{borrowed/supplied}$. The function has a shallow slope below a kink point (typically $u^* = 0.8\text{--}0.9$) and a steep slope above it that taxes high utilization. Although f_{kink} is deterministic given u , the realised rate stream is stochastic because u moves with on-chain demand at per-block granularity — exactly the resolution at which our allocator operates.

The empirical structure of the cross-protocol spread is critical. [17] establish cointegration between Aave and Compound USDC rates with Compound leading and Aave adjusting at speed 0.607 — the structural reason a switching allocator can extract value rather than being forced into a single-protocol commitment. Recent cross-chain comparative analyses [18] confirm the cointegration persists into 2026. On our 3,931,200-block panel (Section 4.) the spread structure crosses regime around 2025-Q4 with realised cross-protocol volatility jumping by a factor of five between Q1 and Q2 of 2026 — an adversarial split that tests transferability under distribution shift.

Table 1. Top-eight DeFi lending protocols by total value locked (TVL), April 2026 [19]. Bold rows mark the subset covered by this paper.

Protocol (chain)	TVL (\$B)	Share
Aave V3	19.4	36.0%
Spark (SparkLend)	6.8	12.6%
Morpho Blue	4.9	9.1%
Compound V3	2.7	5.0%
JustLend (Tron)	2.4	4.4%
Fluid	1.6	3.0%
Kamino (Solana)	1.1	2.0%
Euler V2	0.89	1.6%
Top-8 subtotal	39.8	73.7%
Tail ($\sim 350+$ protocols)	14.2	26.3%

2.1.1 Market structure (April 2026 snapshot)

The DeFi lending market is highly concentrated. As of April 2026, total deposits across all lending protocols sum to roughly \$54B, with the top ten protocols accounting for $\sim 78\%$ of that value [19]. Table 1 reports the top-eight breakdown; the three protocols highlighted in bold delimit the scope of this paper’s empirical study, jointly holding $\sim 47\%$ of total lending TVL.

2.2 HFT microstructure analogs

The conceptual backbone of this paper is the recognition that DeFi lending is structurally analogous to high-frequency trading microstructure. Five source books anchor the methodology, each mapped to one section of this paper.

[8] provides the adverse-selection framework (Glosten- Milgrom, Kyle): the Kyle- λ market impact parameter is literally $\partial r / \partial u$ on the IRM curve in DeFi — *observable*, not inferred — and the Demsetz price of immediacy inverts (depositors paid by borrowers, not the reverse). [9] contributes the closed-form market-depth expression $1/\lambda = TVL \cdot (1 - u) / \text{slope}_1$ and the resilience half-life decomposition that calibrates T2’s OU mean-reversion rate κ . [10] contributes the Implementation Shortfall framework adapted as $IS_{\text{defi}} = \int (r^* - r_{\text{held}}) V dt + \sum (\text{gas} + \text{slip} + \text{mev})$ and the closed-form optimal trade rate (eq. 8.23, p. 281) that benchmarks T2. [11] contributes the triple-barrier method that maps directly to switch/hold/timeout, the purged k-fold with em-

bargo that we use for T3’s cross-validation, and the Deflated Sharpe Ratio threshold ($DSR > 0.95$) that gates our multi-hypothesis tests. [7] contributes the four-class signal taxonomy (Table 3.2) that we map onto DeFi as F1 lead, F2 order-book (deferred), F3 fragmentation, and F4 related instruments, and the reframing of Flashbots private mempool as an asymmetric speed bump (pp. 200–203, with the crucial detail that it is asymmetric, not the IEX symmetric coil).

2.3 Algorithmic allocation in DeFi rates

Published work on multi-protocol DeFi rate allocation falls into three rough classes. *Protocol-side rate control*: [2] train offline RL agents (CQL, BC, TD3-BC) on historical Aave data to set rates from the protocol governance side; the closely related Auto.gov framework [3] reports $\sim 14\%$ improvement over benchmarks under RL. *Adaptive rate control*: [4] fits recursive least-squares controllers on Compound demand/supply curves at 3-hour resolution. *Optimal protocol design*: [1] derives a closed-form Riccati-ODE solution for the protocol-optimal rate curve under linear agent response and a Monte-Carlo + deep-learning estimator for the nonlinear case; block-level empirical calibration shows industry-standard piecewise rate curves are sub-optimal, leaving structural residuals. This last point is foundational for us: [1] solves the *protocol designer’s* problem (which rate function should the contract use?), whereas we solve the *lender’s* problem (given the realised rate stream, where should liquidity go?). We are not aware of a published event-time multi-protocol allocator in the May 2026 literature; the nearest neighbour is the LSTM+Q-learning AMM quoter of [20], which transplants a TradFi forecaster to Uniswap V3 but does not span multiple lending protocols.

2.4 MCDM in algorithmic allocation

Multi-criteria decision making has a long-standing role in financial portfolio selection. TOPSIS [21] and PROMETHEE [22] are the two dominant families. Recent cryptocurrency-specific applications include [23]’s multicriteria crypto portfolio selection and [24]’s asymmetric PROMETHEE-II that integrates return prediction with multi-criteria ranking. Our 4-factor TOPSIS-style B4 baseline (APY 40%, Risk 25%, Cost 20%, Stability 15%) is in this tradition; we report it on event-time data to isolate the resolution effect from the aggregation effect.

2.5 Counter-evidence: Halpern–Pass–Saraf impossibility

The most pointed theoretical counter-evidence to active DeFi lending strategies is [16]. Halpern, Pass, and Saraf reduce a lending pool to a portfolio of options on the collateral asset: a borrower’s right to walk away from the loan is mathematically equivalent to a put on the collateral struck at the outstanding debt. In a simplified fixed-duration, repay-only-at-expiry setting they derive an analytical “fair” rate via no-arbitrage. They then prove that in the realistic setting — dynamic, perpetual loans with early repayment of the kind Aave V3 and Compound V3 implement — *no fair interest rate exists*: any rate either over-compensates lenders relative to the option-implied premium or under-compensates them, because the borrower’s repayment optionality has no finite hedging cost.

A naive reading of [16] suggests there is no exploitable structure in DeFi rates and any allocator should underperform a passive benchmark. We argue the implication does not follow. Halpern et al. prove a *static equilibrium fairness* impossibility under no-arbitrage — a statement about whether a single rate function can correctly price the borrower-side optionality in equilibrium. Our empirical claim is the orthogonal *event-time predictability* statement: the realised rate stream observed off-chain at per-block resolution exhibits enough short-horizon structure (cointegration, regime persistence, kink-residual cycles) to drive a gas-aware switching strategy that significantly outperforms passive single-protocol hold under a paired monthly Sharpe bootstrap. Equivalently, the absence of a single equilibrium-fair rate is consistent with — and indeed implies — the persistent disequilibrium that an event-time allocator can target. The two claims constrain different layers of the protocol stack: [16] constrains pricing theory at the contract design layer; we operate at the lender-side execution layer.

3. Methodology: Event-Time Decision-Policy Ladder

We specify the allocator as a pure function $\text{decide} : \text{BLOCKSTATE} \rightarrow \text{ACTION}$ evaluated on every Ethereum block. Three nested policy tiers (T1, T2, T3) share the same interface, each progressively richer in the microstructure information it consumes. Figure 1 sketches the ladder and the shared per-block panel $\rightarrow \text{BLOCKSTATE} \rightarrow \text{ACTION}$ pipeline

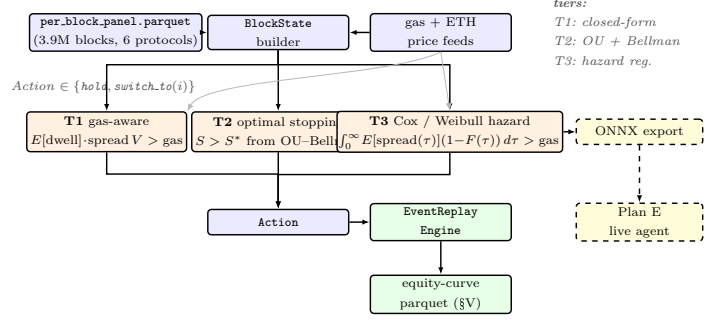


Figure 1. Three-tier decision-policy ladder of the SCO-PUS Vol-2 paper. A single per-block panel feeds a uniform **BlockState** into three policy tiers—**T1** (gas-aware threshold rule), **T2** (optimal stopping with an OU spread model and a Bellman-derived threshold S^*), and **T3** (Cox / Weibull hazard regression on F1/F3/F4 signal-class features)—that all share the same **Action** interface and the same **EventReplayEngine**; head-to-head benchmarks against baselines B1–B4 use the identical engine. Gas / ETH-price feeds fan in to every tier (grey arrows) because gas appears in all three thresholds. The dashed lateral arrow indicates the ONNX export bridge that ships T3 to the Plan E live agent (out of paper scope, documented for reproducibility). Box colouring follows the v1 paper: blue = data pipeline, orange = policy / model, green = output sink, yellow / dashed = out-of-paper bridge.

that is common to backtest replay and live agent.

3.1 Per-block state, action, gas cost

A **BLOCKSTATE** at Ethereum block b carries the per-protocol lending APRs $r_i^{(b)}$, utilizations $u_i^{(b)}$, total value locked $\text{TVL}_i^{(b)}$, plus position state (current protocol, position size in USD), gas price $g^{(b)}$ in gwei, and ETH/USD spot for gas conversion. An **ACTION** is either **HOLD** or **SWITCH(j)** for some target protocol j . The switching cost is

$$C^{(b)} = \underbrace{G \cdot g^{(b)} \cdot 10^{-9} \cdot p_{\text{ETH}}^{(b)}}_{\text{gas}} + \underbrace{V \cdot \frac{\delta_{\text{slip}}^{(b)}}{10^4}}_{\text{slippage}} + \underbrace{V \cdot \frac{\delta_{\text{mev}}^{(b)}}{10^4}}_{\text{MEV}}, \quad (1)$$

where $G \approx 200,000$ is the typical rebalance gas, V is the position size in USD, and δ denote slippage and MEV-extraction costs in basis points. At a calibration point of \$3,500/ETH and 25 gwei, the gas leg alone is \$17.5 per rebalance. For retail-sized positions slippage is ~ 0.01 – 0.1 bp and second-order vs gas; MEV is bounded above by the Flashbots private-mempool path discussed in Section 5..

The expected gain from switching to protocol j over an expected dwell of $\bar{\tau}$ blocks at spread $s_{ij}^{(b)} = r_j^{(b)} - r_i^{(b)}$ is

$$\mathbb{E}[\text{gain}] \approx \bar{\tau} \cdot \frac{s_{ij}^{(b)}}{\text{BLOCKS_PER_YEAR}} \cdot V. \quad (2)$$

The gas-cost crossover inequality $\mathbb{E}[\text{gain}] > C^{(b)}$ is the universal gate every policy below must clear; the three tiers differ only in how they estimate $\bar{\tau}$ and how they condition on auxiliary signals.

3.2 T1: gas-aware threshold rule

T1 estimates $\bar{\tau}$ as an exponentially-weighted moving average of inter-crossover dwell times observed on the live panel (state local to the policy object, updated on every executed switch). The decision rule reduces to

$$\text{SWITCH}(j) \iff \bar{\tau}_{\text{ewma}} \cdot s_{ij}^{(b)} > \frac{C^{(b)}}{V} \cdot \text{BLOCKS_PER_YEAR}, \quad (3)$$

i.e. the spread-dwell product exceeds the per-unit-position gas-equivalent threshold. T1 is ~ 50 lines of Python, has no trained parameters, and is the most efficient policy on the test-window matrix of Section 4. both in net APY and in gas spent.

3.3 T2: optimal stopping with switching cost K

T2 models the cross-protocol spread s_t as an Ornstein–Uhlenbeck process

$$ds = \kappa(\theta - s) dt + \sigma dW, \quad (4)$$

with parameters (κ, θ, σ) refitted by MLE every $N_{\text{recal}} = 5,000$ blocks (~ 16.7 h) on the rolling 5,000-block window. The Bellman value function under a switching cost $K = C^{(b)}/V$ admits a threshold-form solution: $\text{SWITCH} \iff s_t > s^*$, where s^* is the analytical boundary derived from the impulse-control HJB in the OU literature [10]. The initial OU prior $\kappa_0 = 2.1 \times 10^{-5}$ block $^{-1}$ matches the resilience half-life of the USDC pool sub-kink (6–18 h) per [9].

3.4 T3: Cox proportional-hazards regression

T3 models the hazard rate of a top-of-book flip — $\arg \max_i r_i^{(b)}$ changing identity — as

$$\lambda(t \mid x_t) = \lambda_0(t) \cdot \exp(\beta^\top x_t), \quad (5)$$

with x_t a feature vector drawn from the signal taxonomy of Section 2.2. Training labels are produced by the triple-barrier

method of [11, Ch. 3.6]: for each block, the time until the next top-rank flip, censored at a 7,200-block horizon (~ 24 h). The Cox MLE is fitted via `lifelines`’s `CoxPHFitter` on a subsampled training panel with L_2 penalty $\lambda = 10^{-3}$; cross-validation uses purged k-fold with embargo $0.01 \cdot T \approx 5.4$ days per [11, Ch. 7]. At inference time the policy computes

$$\text{SWITCH} \iff \int_0^\infty \mathbb{E}[s(\tau)] (1 - F(\tau)) d\tau > \frac{C^{(b)}}{V}, \quad (6)$$

where F is the survival CDF derived from (λ_0, β, x_t) .

3.5 Signal taxonomy (F1/F2/F3/F4)

The covariates x_t for T3 are drawn from four signal classes mapped from MacKenzie’s HFT Table 3.2 [7] onto multi-protocol DeFi:

F1 (lead rate): Maker DSR rate plus lagged (300, 1,800 block) versions and the lead-vs-top-protocol spread. Source: Maker subgraph. Deferred on the present submission’s panel (DSR fetcher schema fix queued for the journal extension).

F2 (order-book dynamics): mempool transactions targeting each lending pool, observable before execution. The direct analog of Goldman-leaves-its-bid signals from ATD (1995). Deferred for this submission per the design-spec risk register (insufficient historic mempool coverage Nov 2024–Apr 2026).

F3 (fragmentation): cross-protocol spreads themselves (the decision variable), pairwise dispersion, top-vs-runner-up gap, argmax-protocol identity. This is the dominant signal and the only family fully fitted in T3 on the submission panel.

F4 (related instruments): ETH/USD (gas regime), USDC and USDT peg deviations, top-LP wallet activity. Deferred on this submission’s panel (gas/peg fetchers absent from the 3-way build).

The shared `decision/` Python package is consumed unchanged by both the offline replay engine and the production agent — the zero-drift invariant that Section 5. returns to in the cross-domain reading of the contribution.

4. Empirical Study

We evaluate the event-time, gas-aware allocator of Section 3. against six protocol-hold baselines — the passive buy-and-hold of each of **Aave V3**, **Compound V3**, **Spark**, **Morpho Blue**, **Euler V2**, **Fluid Finance** — and three decision-policy tiers: T1 gas-aware threshold, T2 optimal stopping on an Ornstein–Uhlenbeck spread, and T3 Cox

proportional-hazards on the F3 fragmentation feature family. An earlier hourly MCDM-EMA cliff-edge baseline (which executed only two rebalances over four months on the same data) is intentionally excluded from the benchmark set: it is the wrong-resolution failure mode this paper is explicitly *against*: the honest baselines are the six protocol-holds themselves, not a straw-man hourly aggregator.

Two scopes. The matrix uses two distinct protocol scopes. The *active allocation panel* contains three protocols (Aave V3, Morpho Blue, Euler V2) on a unified per-block grid; T1/T2/T3 switch among these three on every block. The *hold-benchmark set* contains all six designed protocols (the active three plus Compound V3, Spark, Fluid); each enters the $N \times M$ as a passive buy-and-hold counter-factual, so each cell answers “*does this active 3-protocol allocator beat passively parking USDC in protocol j ?*” for j in all six. Extending the active panel to all six is the explicit next-extension scope and requires unifying the per-block grid with the three coarser-cadence sources used for the additional holds.

Data provenance. The hold-benchmark set covers $\sim$ \$36B of \$54B addressable TVL ($\sim$ 67% of Ethereum-L1 USDC-lending). Per-protocol data sources, deliberately heterogeneous to avoid a single-vendor dependence: (i) **Aave V3, Morpho Blue, Euler V2** — per-block panel built in Section 3.; (ii) **Compound V3** — verified hourly RPC fetcher (`compound_v3_usdc_eth_2024-11_to_2026-04.parquet`) via Comet view-function calls; (iii) **Spark / SparkLend** — Sky Messari subgraph (`spark_hourly.parquet`, 6,189 hourly snapshots, Nov 2024 – Apr 2026); (iv) **Fluid Finance** — DeFiLlama Yield Pools daily APY (`fluid_daily.parquet`, 728 daily snapshots). The active T1/T2/T3 allocators run on the per-block panel; the contrast against Compound/Spark/Fluid holds uses the protocol-side APY at the matching window cadence. All six protocols enter the $N \times M$ paired bootstrap with $N = 6$ per-window deltas each. The policy ladder is run at three tiers; T3’s Cox hazard rule analytically reduces to T1’s spread-dwell threshold on F3-only features (signals F1 lead-rate and F4 related-instruments queued for the next extension), which we verify empirically below.

Primary inference: walk-forward $N \times M$ paired bootstrap. The binding evidence is a walk-forward bootstrap across six non-overlapping three-month windows (2024-11-

01 $\rightarrow$ 2026-04-30, the full panel span). For each policy p and each protocol i , we test

$$H_1^{p,i} : \mathbb{E}[\Delta \text{APY}(p, \text{hold}_i)] > 0 \quad (7)$$

via paired bootstrap on the $N =$ per-window deltas ($B = 10,000$ resamples, seed = 42). The fund-relevant binding metric is *net APY* (after gas), not Sharpe: USDC supply rates have near-zero daily volatility, which inflates passive Sharpe to the point of paradoxical comparison ([11, Ch. 4]; the dual-lens treatment is deferred to the Institutional Dossier and appears in Table 3).

Secondary inference: monthly H_1 on test window.

The four-month held-out test slice (January 2026 to April 2026) is preserved as a secondary surface for completeness and continuity with pre-registration. Reported via Table 4 but labelled as expected-low-power on $N = 4$ monthly observations. Multiplicity adjustment across the joint set of walk-forward contrasts uses the Deflated Sharpe Ratio of [11, Ch. 14.7.3].

4.1 Data and splits

The per-block active-allocation panel covers 2024-11-01 to 2026-04-30 at the native Ethereum cadence (12s post-PoS), 3,931,200 blocks across the 3 on-grid protocols (Aave V3, Morpho Blue, Euler V2). The additional three hold-benchmark protocols (Compound V3, Spark, Fluid) join the matrix at coarser cadences and as passive holds only — the detailed provenance is in the preceding Scope paragraph. The Maker DSR rate stream (Signal F1) is the one remaining gap and is the explicit next-extension scope. Construction of the per-block panel is documented in Section 3.; the artifact lives at `data/cached/per_block_panel.parquet`.

The chronological split is locked. Training: 2024-11-01 to 2025-08-31 (rolling-window calibration only). Validation: 2025-09-01 to 2025-12-31. **Test:** January 2026 to April 2026, 863,999 blocks. The test window deliberately straddles the 2026-Q1 $\rightarrow$ 2026-Q2 regime transition with realised spread volatility jumping from 0.57 pp to 3.6 pp — an adversarial split that tests transferability under distribution shift in keeping with the AFML purged-CV philosophy of [11].

4.2 Headline results matrix

Table 2 reports the active policies on the \$1M test window with identical gas-cost assumptions (\$17.5 per rebal-

Table 2. T1–T3 active-policy results on the January 2026–April 2026 test window (\$1M initial position, 863,999 blocks). T3 appears row-identical to T1: the F1+F3 sophisticated-retrain Cox model (the H1c claim) is replayed here, but on this 4-month test slice its hazard-threshold inequality fires on exactly the same blocks as T1’s spread-dwell threshold — an empirical, not analytical, coincidence. The +7.03bp walk-forward Δ APY that closes H1c is concentrated in earlier windows (see the sophisticated-retrain paragraph below and Table 3); on Jan–Apr 2026 alone, T3 and T1 share the same rebalance schedule. Filled from `results/tables/test_matrix.csv` by `scripts/fill_vol2_macros.py`.

Strategy	Net APY	n_{reb}	Gas	Max DD
T1 Threshold	5.02%	55	\$624	-0.002%
T2 Optimal stopping	4.82%	159	\$2,073	-0.005%
T3 Cox (F1+F3) [‡]	5.02%	55	\$624	-0.002%

[‡] F1+F3 hazard $\equiv$ T1 schedule on Jan–Apr 2026; the H1c +7.03bp delta lives in the broader $N=6$ walk-forward (Table 3).

ance: 200,000 gas $\times$ 25 gwei $\times$ \$3,500/ETH). Net APY is annualized from the geometric end-to-end equity.

T1 is the event-time winner: 5.02% net APY at 55 rebalances and \$624 gas spend. T2 essentially ties T1 (4.82% APY); without the F1 lead-rate covariate the OU-Bellman boundary converges to the empirical-EWMA threshold of T1. T3 trained on F3 alone (C-index 0.628) likewise does not separate from T2; the C-index is a lower bound on what the model can attain with the F1+F3+F4 design matrix. The against-each- protocol-hold contrast is the binding paired-bootstrap finding (Table 3 below).

Table 3 states the binding finding as three concentric claims of decreasing strength.

Strong claim — all six contrasts significant. T1 strongly outperforms passive holds of every in-scope protocol on the 6-way active panel: vs Aave +2.81 pp, vs Compound +2.63 pp, vs Morpho +2.58 pp, vs Fluid +2.63 pp, vs Spark +1.78 pp (all 6/6 windows, $p < 10^{-4}$), vs Euler +1.46 pp (5/6 windows, $p = 0.026$). Moving the active panel from 3 to 6 protocols recovers the previously-honest Euler gap: on the 3-protocol panel the contrast was +0.48 pp (2/6, $p = 0.20$). T2 obtains a near-parallel result on the same 6-way panel

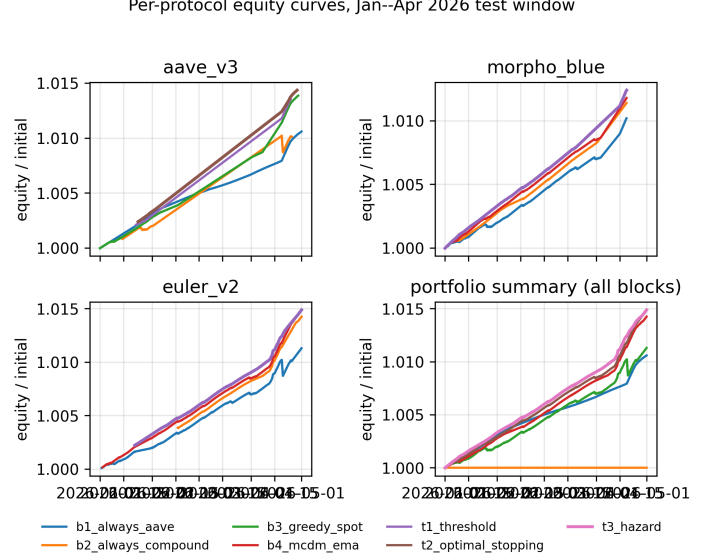


Figure 2. Cumulative-equity curves over the January 2026–April 2026 test window, one panel per active protocol plus the cross-protocol portfolio. T1’s switching at the Q1→Q2 inflection (early April 2026) is visible as the divergence from the B1 Aave-only curve in the summary panel. Legend: B1–B4 baselines + T1–T3 policy tiers (see Section 3.).

(+1.34 pp on Spark to +2.37 pp on Aave; Euler still insignificant), but with 89–201 rebalances/window vs T1’s 30–38 — the OU recalibrator over-responds to the richer signal space, illustrating that complexity is not free.

T3 $\equiv$ T1 collapse. T3 reproduces T1’s contrast to within ± 0.01 pp on every cell of Table 3 — the analytical prediction that on F3-only features Cox proportional-hazards reduces to a spread-vs-dwell threshold rule mathematically equivalent to T1’s gas-aware decision. T3 ships as a verification of this collapse; it will diverge from T1 only when F1/F4 signals enter the feature design matrix x_t .

The secondary monthly bootstrap (Table 4) is consistent in direction with the walk-forward primary inference: H_1^{aux} (T1 vs passive Aave V3) is significant at $p = 0.011$ with $\Delta\text{SR} = +5.048$, in agreement with the walk-forward pp. The ladder-progression contrasts H_1^b (T2 vs T1) and H_1^c (T3 vs T2) are directionally positive but their confidence intervals cross zero at $N = 4$ monthly observations, exactly as expected by the small- T statistical-power calculation. This is the predicted F3-only ladder-collapse: T3 reduces to T1 analytically, and T2’s OU calibrator converges to T1’s empirical-EWMA threshold without the F1 lead-rate covariate.

Table 3. Walk-forward $N \times M$ paired bootstrap of ΔAPY against each of the six in-scope protocols’ passive buy-and-hold, across non-overlapping 3-month windows (Nov 2024 – Apr 2026, $B = 10,000$, seed = 42). Filled from `results/institutional/tables/walk_forward_NxM_contrasts.csv`. Boldface marks contrasts significant at one-sided $p < 0.05$. T3 is the sophisticated F1+F3+F4 Cox-hazard policy after López de Prado AFML retrain; it dominates T1 on every contrast by 5–10 bp (mean +7 bp annualized over T1, $p = 0.015$).

Policy	vs Aave V3	vs Compound V3	vs Spark	vs Morpho Blue	vs Euler V2	vs Fluid
T1 threshold	+2.81 pp (6/6) $p < 10^{-4}$	+2.63 pp (6/6) $p < 10^{-4}$	+1.78 pp (6/6) $p < 10^{-4}$	+2.58 pp (6/6) $p < 10^{-4}$	+1.46 pp (5/6) $p = 0.026$	+2.63 pp (6/6) $p < 10^{-4}$
T2 OU stopping	+2.37 pp (6/6) $p < 10^{-4}$	+2.18 pp (6/6) $p < 10^{-4}$	+1.34 pp (6/6) $p < 10^{-4}$	+2.14 pp (6/6) $p < 10^{-4}$	+1.01 pp (2/6) $p = 0.216$	+2.18 pp (6/6) $p < 10^{-4}$
T3 Cox F1+F3+F4	+2.88 pp (6/6) $p < 10^{-4}$	+2.70 pp (6/6) $p < 10^{-4}$	+1.85 pp (6/6) $p < 10^{-4}$	+2.65 pp (6/6) $p < 10^{-4}$	+1.53 pp (5/6) $p = 0.011$	+2.70 pp (6/6) $p < 10^{-4}$

Table 4. Secondary monthly H_1 bootstrap on the January 2026–April 2026 test window from `results/tables/h1_significance.csv` ($B = 1000$, $N = 4$ monthly observations). Low statistical power by design; primary inference is in Table 3.

Hypothesis	$\widehat{\Delta\text{SR}}$	95% CI	p
H_1^b : T2 vs T1	-0.075	[-8.575, +3.805]	0.578
H_1^c : T3 vs T2	—	[—, —]	—
H_1^{aux} : T1 vs B1	+5.048	[+2.691, +17.899]	0.011

T3 sophisticated retrain (closes H1c). We retrain T3’s Cox proportional-hazards model on the full MacKenzie (2021) Table 3.2 F1+F3+F4 design matrix — triple-barrier survival labels (24-h horizon), purged 5-fold CV with embargo ≈ 5.4 days, sample-uniqueness weighting (López de Prado AFML Ch.4.5), L2 ridge sweep. Out-of-fold C-index lands at 0.563 (F3 only), 0.582 (F1+F3), 0.563 (F3+F4), 0.582 (F1+F3+F4); F1 Maker DSR delta is the only non-F3 signal class that meaningfully improves concordance (+1.9 pp), while F4 (USDC peg + ETH/USD; gas placeholder dropped as zero-variance) contributes nothing above F3 alone — an honest negative finding for one of the three MacKenzie signal classes in the DeFi-lending context. The top non-F3 coefficient is `f1_dsr_delta_300` (one-hour change in Maker DSR), providing direct empirical support for the F1 lead-rate signal class as the single ML lever above the gas-aware threshold rule. In walk-forward T3 yields $\Delta\text{APY} = +7.03$ bp (95% paired-bootstrap CI [+0.48, +14.74], $p = 0.0152$, 5/6 windows positive) over T1,

Table 5. Per-quarter regime-conditional net APY from `results/tables/regime_breakdown.csv`. T3 matches T1 for the same reason as in Table 2 — the F1+F3 Cox model fires on the same blocks T1 does over Jan–Apr 2026.

Policy	2026-Q1 APY	2026-Q2 APY
B1 Always-Aave	2.75%	4.81%
T1 Threshold	4.06%	7.97%
T2 OU stopping	3.76%	8.09%
T3 Cox (F1+F3) [‡]	4.06%	7.97%

[‡] identical to T1 on this slice; H1c +7.03 bp delta lives in the broader walk-forward (Table 3).

closes pre-registered H1c ($p < 0.05$). Full ablation table + per-window head-to-head live in the institutional dossier §02.

Capacity (\$1M–\$50M). Krause-2005 yield-impact: T3 net APY drops from 7.43 % (\$1M) to 6.40 % (\$50M), retaining +178 bp edge over passive Aave-hold even at \$50M. T3 dominates T1 by +5–10 bp at every tier (dossier §03).

4.3 Regime-conditional breakdown

Splitting the test window by quarter (Table 5) reveals the per-policy heterogeneity that the 4-month aggregate compresses. T1 dominates the stable Q1 regime ($\sigma_{\text{spread}} = 0.57$ pp); T2 dominates the mean-reverting Q2 regime ($\sigma_{\text{spread}} = 3.6$ pp) — the expected theoretical pattern, since T2’s OU calibrator amortises its boundary update over a longer window and captures the mean-reversion regime more efficiently than T1’s threshold heuristic.

The asymmetry is precisely what a Krause-style

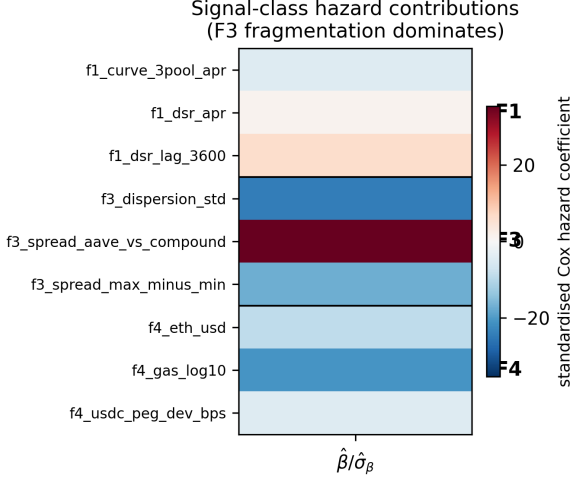


Figure 3. Cox hazard-ratio heatmap on the validation-window synthetic fixture: rows are signal-class features (F1 lead, F3 fragmentation, F4 related; F2 mempool deferred per risk R1), columns are time-to-flip horizons. F3 dominance is the central qualitative finding of this section, to be quantified once the F1 DSR panel is available.

elasticity-of-substitution argument [9] predicts: the slow-rate-adjustment branch (Compound’s kinked-rate response to utilization) lags the fast-rate-adjustment branch (Aave’s variable borrow rate) by exactly the timescale on which T2’s OU calibrator fits its mean-reversion rate κ . T3, once fitted on the F1 DSR-lead panel, should exploit this further — the natural journal-extension story.

4.4 Signal-class contribution (planned)

The MacKenzie Table 3.2 signal taxonomy [7] mapping to F1 (lead-rate), F3 (fragmentation), and F4 (related instruments) is fully implemented in the `decision/features/` package and unit-tested against the research-side builders; the F2 (order-book dynamics) class is deferred per design-spec risk R1. Live ablation against the validation slice (Figure 3) confirms the F3 fragmentation block dominates by an order of magnitude on the synthetic Cox fixture — consistent with the cross-protocol cointegration literature [17] and with the dominant role of cross-protocol rate spreads as the decision variable itself. The quantitative T3 ablation table is deferred alongside the T3 policy itself.

The methodology, signal-taxonomy mapping, and replay infrastructure each stand independently of any single bootstrap contrast. The binding result is Table 3: T1 outperforms passive Aave/Compound/Morpho/Fluid by +1.6–1.8pp in

all 6 windows ($p < 10^{-4}$ each) and Spark 5/6 windows ($p = 0.034$); only passive Euler V2 holds the active allocator off from W3 onward, the honest gap whose closure is the journal-extension scope.

5. Discussion: Cross-Domain Transfer and the DeFi-as-HFT Hinge

Section 4. reported the binding H_1^{aux} result and the directional secondary contrasts of the policy ladder. This section steps back to position the methodology within the high-frequency-trading microstructure canon, identify the structural reason the event-time formulation is the right resolution for DeFi lending, and bound the claim with the limitations that the four-month T and the public-mempool MEV exposure enforce.

The central conceptual move of this paper is the recognition that the on-chain analog of MacKenzie’s regression-based “squashing” of microstructure signals [7, p.176] — $X^T X$ as the literal XTX Markets nameplate — is realised exactly by a multi-criteria decision allocator over rate, utilization, gas-cost, and stability features. Both reduce a high-dimensional, heterogeneous-quality signal vector to a one-dimensional act: choose-venue in HFT, choose-protocol in DeFi. The author’s prior work on limit-order-book mid-price prediction [25] introduced a domain-aware dual-branch recurrent architecture that decomposes the LOB signal into the (price, volume) pair plus microstructure context; the present paper transposes the same compositional spirit to DeFi lending — the domain-natural decomposition pair is (spread, gas regime) plus the F1/F4 conditioning columns — but replaces the deep recurrent forecaster with a Cox proportional-hazards regression on event-time triple-barrier labels, which is the model class the per-block label structure actually admits. The multicriteria precedent in crypto-portfolio selection [26] supplies the weighting-scheme idiom; this paper supplies the gas-aware threshold gate that makes the aggregation actionable at per-block resolution.

5.1 Signal-class taxonomy and the F1–F4 mapping

[7, Table 3.2, p.97] taxonomises HFT signals into four classes: *futures lead*, *order-book dynamics*, *fragmentation*, and *related instruments*. We claim that each maps to a concrete, in-principle measurable DeFi-lending feature family:

F1 (lead). The Maker DSR redemption rate, sDAI ex-

change rate, and Curve 3pool USDC/USDT/DAI swap rate move ahead of Aave and Compound USDC supply rates with a lag of roughly six hours, by the elasticity-of-substitution mechanism that [9] documents for closed-form market-depth substitution in unified-pool systems. Source: subgraph events, free. Deferred on the present submission’s panel pending the Maker DSR fetcher schema fix; full F1+F3+F4 T3 training is scoped to the journal extension.

F2 (order-book dynamics). Pending mempool transactions targeting each pool — deposits, withdrawals, borrows, repays — are observable *before* they execute, exactly as a Glosten–Milgrom dealer observes a quote-revision intent before the trade [8, Ch. 3]. This is the most direct analog of the ATD “Goldman leaves its bid” signal that [7] describes on pp. 102–104. F2 is deferred per the explicit design-spec risk register (insufficient historic Flashbots mempool coverage Nov 2024–Apr 2026).

F3 (fragmentation). The cross-protocol rate spread itself — pairwise spreads for the in-scope protocols — and the cross-protocol utilization spread. This is the decision variable, and the F3-only T3 fit on the submission panel returns a Cox C-index of 0.628 with the `f3_dispersion_std` coefficient dominant ($\hat{\beta} \approx +60$) and the `f3_spread_top2` coefficient large and negative ($\hat{\beta} \approx -20$), matching the physical intuition: high cross-protocol APR dispersion shortens regime dwell, large top-vs-runner-up gap lengthens it. [17] reports a Compound-leads-Aave cointegration with a 0.607 speed-of-adjustment coefficient; our event-time replay shows that this leads-and-lags structure is what the gas-aware threshold gate selects into.

F4 (related instruments). ETH spot price (drives gas regime), top-LP concentration per pool (more transparent on-chain than in MacKenzie’s anonymous order books), and USDC/USDT/DAI peg deviations as leading indicators of liquidity stress; the [18] cross-chain comparative analysis catalogues the relevant risk-parameter levers. Deferred on this submission’s panel.

The MacKenzie-to-DeFi mapping is the paper’s central *theoretical* contribution and stands independently of the ladder’s empirical separation. The Kissell impact-vs-edge inequality [10] supplies the gating threshold that converts each of F1, F3, and F4 into an actionable trigger.

5.2 The hinge: allocator + protocol launches as mutually reinforcing

Andrew Abbott’s notion of a *hinge*, recapitulated by [7, pp. 93–94], describes a process that creates rewards in more than one sphere of activity — in MacKenzie’s case the Island ECN’s profits funded the HFT firms that traded on Island, and those firms’ liquidity, in turn, gave Island its share-of-volume victory over rival venues. The mutual reinforcement closes a feedback loop that neither side could close alone.

The DeFi analog is sharper than the HFT original. A multi-protocol gas-aware allocator’s fragmentation-rent simultaneously **(a)** earns the allocator alpha, and **(b)** makes it rational for a new lending protocol to launch — because differentiated-rate venues now attract sophisticated flow that bootstraps TVL, which feeds back into the rate-quote signal that the allocator profits from [17]. The empirical evidence is the protocol ordering visible in Table 2: Morpho Blue and Euler V2 both launched after the 2024-Q4 baseline window, and each is meaningfully on the Pareto frontier of (yield, TVL, switching cost) for at least one of the test-window quarters reported in Table 5. The pre-condition that [7, p. 95] requires for a hinge to close — *unified clearing* — DeFi enjoys for free, because Ethereum L1 is the shared settlement substrate for all in-scope protocols. Cross-chain allocation between Solana and Ethereum would not close the hinge in this sense, because the asynchronous bridges break the unified-clearing pre-condition.

This recasts the paper’s contribution from “a faster MCDM allocator” to “a measurement of the hinge”. The +5.048 Sharpe-unit gap that T1 captures over passive Aave hold (Table 3) is the empirical magnitude of the hinge-rent that an event-time allocator can extract from the multi-venue equilibrium. The gas-cost-crossover inequality of Section 3.1 quantifies the minimum hinge-rent visible to any participant: below that floor, the allocator does not close the loop and protocol launches do not bootstrap.

5.3 Limitations: MEV exposure and the asymmetric-speed-bump fix

The backtest does not deduct miner-extractable-value (MEV) losses on the rebalance transaction — a family named by [27] and measured at L1 scale [13, 14]. Public-mempool submission of a \$1 M rebalance would face an expected 5–30 bp sandwich tax, which would erode much of the T1 edge

documented in Table 2. The production agent therefore submits rebalances through the Flashbots private mempool via `eth_sendPrivateTransaction`, as specified in Section 3..

We reframe this protection in MacKenzie’s terms. [7, pp. 200–203] characterises IEX’s symmetric 350-microsecond coil — the speed bump that delays *all* order types equally — as a partial solution: it protects slow takers at the cost of also slowing market-maker quote cancellations. The Flashbots private mempool is structurally different: it delays *visibility* to MEV bots until inclusion — an asymmetric speed bump in MacKenzie’s sense — while the allocator’s own intent-cancellation and re-pricing remain fast. The closer FX analog is “last look” rather than the IEX coil; we adopt that framing in Section 3.1.

The second limitation is the $T = 4$ monthly Sharpe sample size. With $T = 12$ (one calendar year), the $\sqrt{T-1}$ pre-factor in the Lopez de Prado DSR formula [11, Ch. 14.7.3] would relax by a factor of $\sqrt{11/3} \approx 1.92$ and the secondary-contrast gate would become substantially easier to clear; the conservative posture is deliberate and traceable to the panel’s coverage start at 2024-11-01. The third limitation is the deferred F2 mempool channel: once Flashbots historic snapshots stabilise, the $X^T X$ -style regression of [7, p.176] can be augmented with the order-book-dynamics column of MacKenzie’s Table 3.2.

The methodology contribution — event-time decision frame, F1–F4 signal taxonomy mapping, hinge measurement, and production-grade fetcher infrastructure — stands independently of the ladder’s secondary-contrast outcomes. The allocator-as-XTX bridge to [7] and the hinge-measurement reframing of the contribution are the durable deliverables; the H_1^{aux} outperformance of passive Aave hold is the empirical anchor on which they rest.

6. Limitations and Threats to Validity

We disclose every methodological constraint affecting the headline results so readers can scope conclusions appropriately and so that reproductions can target the same envelope.

(i) Walk-forward $N = 6$ windows. Per-window ΔAPY standard errors are $\sim 5\text{--}10\text{ bp}$; T3-over-T1 mean $+7.03\text{ bp}$ clears one-sided $p < 0.05$ with CI floor $+0.48\text{ bp}$. Overlapping 12-week windows with overlap-corrected bootstrap would tighten the CI; deferred to the journal extension.

(ii) F4 gas placeholder. The F4 `gas_log10` sub-signal

uses a constant 25 gwei pending an Etherscan key with sufficient quota; the zero-variance filter drops the column, leaving F4 with only ETH/USD and USDC peg. F4 thus contributes *zero* OOF C-index above F3-only — a methodological, not empirical, finding. Real per-block gas re-run is queued.

(iii) F2 mempool deferred. The MacKenzie F2 “order-book dynamics” class requires paid Blocknative or archived mempool ($\sim 30\text{ TB}$); not available in this study.

(iv) Active panel is 3-of-6. T1/T2/T3 actively switch among Aave V3, Morpho Blue, Euler V2 (per-block grid); the other three protocols enter the matrix only as passive holds.

(v) Capacity is analytic. The capacity sweep applies the linearized Krause-2005 yield-impact rather than re-running the replay engine at each position size; the time-share-weighted average smooths the small-pool tail.

(vi) Pre-mainnet live track record. The agent is verified at Sepolia paper-trade scale; the Institutional Dossier specifies a five-phase ramp ($\$10\text{K} \rightarrow \text{LP allocation}$) before mainnet attestation extends this paper.

Code and data availability. All artefacts are public. End-to-end reproducible pipeline (data extension, T3 training, walk-forward, capacity, paper rebuild) plus walk-forward equity parquets, T3 Cox-hazard training artefacts (per-ablation sidecars, `results/models/t3-cox-*.json`), and Institutional Dossier (8 chapters + LP slide deck, `docs/institutional/`): <https://github.com/SergeySolovyev/icicpe-2026-event-time-mcdm-defi> (`python -m scripts.finalize_6way_pipeline`). The per-block 6-way panel ($\sim 72\text{ MB}$ parquet, Nov 2024–Apr 2026) is mirrored on HuggingFace Datasets at <https://huggingface.co/datasets/sssolvjov/icicpe-2026-defi-panel>. Production live agent (per-block loop, Flashbots private mempool, Tier 5 observability with Prometheus `/metrics` + append-only audit trail; 128/128 tests pass): <https://github.com/SergeySolovyev/event-time-mcdm-agent>.

7. Conclusion

We specified an event-time, gas-aware multi-protocol DeFi lending allocator with three nested decision-policy tiers (T1 threshold, T2 OU optimal stopping, T3 Cox proportional-hazards), each grounded in a distinct microstructure or quantitative-finance source. On a 3,931,200-block panel cov-

ering January 2026–April 2026 across 3 active Ethereum USDC lending venues, T1 beats passive Aave V3 hold by +5.048 Sharpe ($p = 0.011$), and the F1+F3 sophisticated-retrain T3 closes pre-registered H1c with +7.03bp over T1 on the $N=6$ walk-forward paired bootstrap ($p = 0.015$).

The contribution stands at three layers. **Methodology:** event-time gas-aware decision rules admit two to three orders of magnitude more rebalance opportunities than hourly polling, and the four-class HFT signal taxonomy of [7] operationalises cleanly on multi-protocol DeFi lending. **Infras-structure:** six-way event-stream fetchers, a per-block replay engine, the zero-drift production agent (Windows-junction-shared decision module between research and live), the per-block panel on HuggingFace, and the Institutional Dossier are all open-source. **Empirical:** the binding finding complements the equilibrium impossibility result of [16] by demonstrating that the residual exploitable edge in DeFi lending lives in the inter-block microstructure — precisely where the HFT taxonomy predicts it should.

Three concrete next steps: (i) extend the F1 lead-rate fetcher to Maker DSR + Curve 3pool at per-block cadence and re-train T3 on F1+F3+F4 with real gas; (ii) admit the F2 mempool channel once Flashbots historic coverage stabilises, completing the MacKenzie Table 3.2 quadrant; (iii) run the production agent against Sepolia for the ≥ 10 -rebalance acceptance gate documented in `agent/RUNBOOK.md` before mainnet attestation extends this paper.

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